System Aim

This project simulates a warehouse sorting robot in MuJoCo. The TurtleBot3 starts in a receiving area, visits three storage bays representing delivery stops, and then drives to a charging station. It uses LiDAR to plan routes around walls and shelves, while wheel odometry predicts its position and heading. When the camera detects the charger's ArUco marker within 3 m and the measurement is accepted, the EKF uses it to correct accumulated pose error. The LQR follows the planned path using the estimated state, and camera measurements support the final docking approach.

Performance is assessed under three conditions: normal operation, added encoder noise, and encoder noise combined with a low-grip floor section.

Project targets

- Visit all three storage bays before approaching the charger, reporting the actual position error at each bay.
- Complete the route with **zero wall or shelf contacts**.
- Achieve **lower position RMSE with camera fusion than wheel odometry alone** on the same noisy drive.
- Confirm docking using the camera and finish within **3 cm** of the intended stand-off position and **3°** of the targets heading.

How does the system do this

The TurtleBot3 begins with a known origin and heading, along with predefined storage goals and the charging-station location. During navigation, it relies on its estimated pose rather than simulator ground truth

Wheel encoders measure wheel rotation, which the EKF uses to predict the robot's position and heading. When an accepted camera measurement of the charger's ArUco marker is available within 3 m, the EKF combines it with the wheel-based prediction to correct accumulated error.

LiDAR detects walls, shelves and free space. The path planner uses these measurements, and the estimated robot pose to build an obstacle map and find a route towards each goal. The LQR uses the estimated position and heading to calculate steering corrections that keep the robot following this path.

The aim is to visit all three storage bays and dock accurately at the charger, while demonstrating that camera and wheel-odometry fusion reduces position-estimation error under encoder noise and disturbance. Simulator ground truth is used only to evaluate performance.

TurtleBot3 state space model

The TurtleBot3 Waffle Pi is modelled as a differential-drive platform. Six short MuJoCo tests with ideal encoders validate the wheel-odometry prediction below.

State and inputs

The odometry test starts at (0,0,0) in a coordinate frame fixed to the robot's starting position. The positive x-axis points in the direction the robot initially face, and the positive y-axis points to its left. At each simulation step, the model uses how much each wheel has rotated since the previous step to estimate the robot's position and heading. The coordinates x and y give the estimated position of the point halfway between the wheels, in meters. The heading θ is measured in radians and increases when the robot turns anticlockwise.

From wheel rotation to robot movement

A wheel rolling through one radian travels one radius. Using the wheel radius r = 0.033 m extracted from the STL files of the TurtleBot3 Waffle Pi left wheel, the distances travelled by the wheels are

$$\Delta sL = r\,\Delta\varphi L$$

$$\Delta sR = r\,\Delta\varphi R$$

TurtleBot's centerpoint travels the average of the two distances $\boldsymbol{\Delta s}$. During a turn the outer wheel follows a longer arc. The difference in arc length equals the wheel separation **b** multiplied by the change in heading. The wheel centres are at +0.144 m and −0.144 m across the axle, giving **b = 0.288 m.**

$$\Delta s = (\Delta sR + \Delta sL)/2$$

$$\Delta\theta = (\Delta sR - \Delta sL)/b$$

Updating the pose

The model treats the motion within one step as a circular arc. Let **h** be half the heading change. The factor **c** converts signed arc length to straight line displacement between the two ends of the arc, directed along the midpoint heading.

$$h = \Delta\theta/2 \qquad c = \sin(h)/h \qquad c = 1 \text{ when } h = 0$$

$$x_{k+1} = x_k + \Delta s\, c \cos(\theta_k + h)$$

$$y_{k+1} = y_k + \Delta s\, c \sin(\theta_k + h)$$

$$\theta_{k+1} = \text{wrap}(\theta_k + \Delta\theta)$$

Wrapping keeps heading between $-\pi$ (inclusive) and $+\pi$ (exclusive). For example, if both wheels rotate forward by 1 rad, each travels 0.033 m. The robot advances 3.3 cm along its current heading with zero heading change.

Assumptions

- The robot moves on a level floor.
- Both wheels have the same constant effective radius,
- The axle is rigid,
- the wheels roll without slipping or sliding sideways.
- Curvature is assumed to be constant within each simulation step.

Motion model validation

Each test started after a settling period. The model accumulated the measured wheel rotations throughout the drive and two seconds of stopping motion. MuJoCo supplied the actual final pose for evaluation in the same starting frame. The test used ideal encoder readings and no camera corrections.

Final pose comparison

Position pairs are (x, y) in metres. Heading error is the signed difference between the predicted and actual headings, wrapped to the shortest angle. Position error is the straight-line distance between the two final positions.

Movement	Predicted (x, y) m	Actual (x, y) m	Heading (°) pred. / actual	Position error (mm)	Heading error (°)
Straight	0.4890, -0.0004	0.4879, -0.0004	-0.097 / -0.096	1.1	0.000
Left turn	-0.0003, -0.0006	-0.0012, -0.0018	112.782 / 112.533	1.5	+0.249
Reverse	-0.1433, -0.0000	-0.1430, -0.0000	0.018 / 0.018	0.3	0.000
Right turn	-0.0003, 0.0006	0.0005, -0.0006	-112.776 / -112.514	1.4	-0.262
Left curve	0.1874, 0.0391	0.1870, 0.0384	23.690 / 23.540	0.9	+0.150
Right curve	0.1874, -0.0391	0.1874, -0.0385	-23.688 / -23.511	0.6	-0.177

What the results show

On the straight drive, the model predicted 0.4890 m forward and MuJoCo recorded 0.4879 m, a difference of 1.1 mm. The reverse drive produced the smallest position error, 0.3 mm. On the spot turns, odometry slightly overestimated rotation in both directions. For the right turn it predicted 112.776° of rotation against 112.514° measured, a difference of 0.262°.

These endpoint comparisons support the wheel-motion model for short manoeuvres under the tested conditions. The EKF prediction and standalone odometry also returned identical poses in all six tests because they use the same wheel increments and motion equations. Accuracy under encoder noise and on the slip plate is assessed separately in the navigation runs.

Controllability of the path tracker

Mathematical check. Before running the simulation, we checked whether steering could correct both the robot's heading and its sideways distance from the path. We entered six forward speeds (0.025–0.160 m/s) into the tracking model. All six gave rank 2, meaning steering can correct both errors. At zero forward speed,

the model gave rank 1 because turning in place changes only the heading. The straight-line tracking model was therefore controllable at all six forward speeds tested.

Linear model and controllability calculation

Let $e = [e_y, e_\theta]^T$, where e_y is signed distance to the left of the active straight segment and $e_\theta = \text{wrap}(\hat{\theta} - \theta_{path})$. For forward speed v, the exact lateral equation is $\dot{e}_y = v \sin(e_\theta)$. Linearising near the path gives $\dot{e}_y = v e_\theta$ and $\dot{e}_\theta = \omega$. Thus $A_c = [[0, v], [0, 0]]$ and $B_c = [[0], [1]]$. The input is commanded yaw rate ω; forward speed is scheduled separately.

With zero-order hold and $T = 0.05$ s, $e_{k+1} = A e_k + B \omega_k$, where $A = [[1, vT], [0, 1]]$ and $B = [[vT^2/2], [T]]$. The controllability matrix is $C = [B, AB] = [[vT^2/2, 3vT^2/2], [T, T]]$. Its determinant is $-vT^3$, so $\text{rank}(C) = 2$ whenever $v \neq 0$. At $v = 0$ it has rank 1: rotation changes heading but cannot immediately remove lateral error.

This result concerns the two-state local tracking model. It does not establish global controllability of the complete robot, feasibility of an obstacle-constrained route, or stability after saturation. The separate ALIGN mode handles large heading errors before the LQR is used.

Straight-line simulation

To demonstrate the correction in MuJoCo, the robot started 5 cm to the left of a 1.5 m straight path, facing parallel to it. The LQR used the pose estimated from the wheel encoders to steer the robot towards the line. Camera corrections added encoder noise and the slip plate were disabled for this baseline test. The simulator's actual position was used only to measure performance. The run ended when the robot estimated it was within 2 cm of the path endpoint.

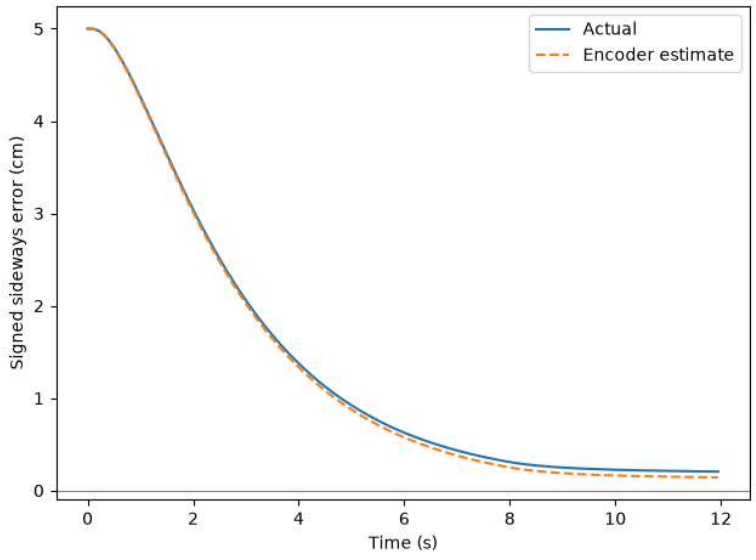


Figure 1. Sideways tracking error during the 1.5 m drive. Zero represents the intended line.

The robot reduced its actual sideways error from 5.00 cm to 0.20 cm (2 mm), correcting 96% of the initial offset without crossing the line. The sideways root mean square error (RMSE) was 2.01 cm over the whole drive, including the initial 5 cm offset. Together, the results show that steering can correct both tracking errors in the model and that the LQR successfully brought the simulated robot close to the intended line.

Observability

Mathematical check. Wheel encoders estimate movement from the known starting position. If the wheels slip, the encoders alone cannot tell how far the robot moved. When the camera sees the known charger marker, it measures range, bearing and heading. At the docking target, the camera measurement matrix had rank 3, meaning these measurements can distinguish changes in all three pose components: x, y and heading.

Simulation result. With encoder noise and the slip plate enabled, wheel odometry alone gave a position RMSE of 17.58 cm. Using the same wheel readings and adding camera corrections reduced this to 8.49 cm, a 52% reduction on this drive. Of the 51 accepted camera updates, 46 immediately improved the estimated position.

Sensing and Calibration

Camera calibration and measurement tuning

The camera was calibrated and its marker corner-refinement settings were tested to support reliable measurements for EKF localisation and docking.

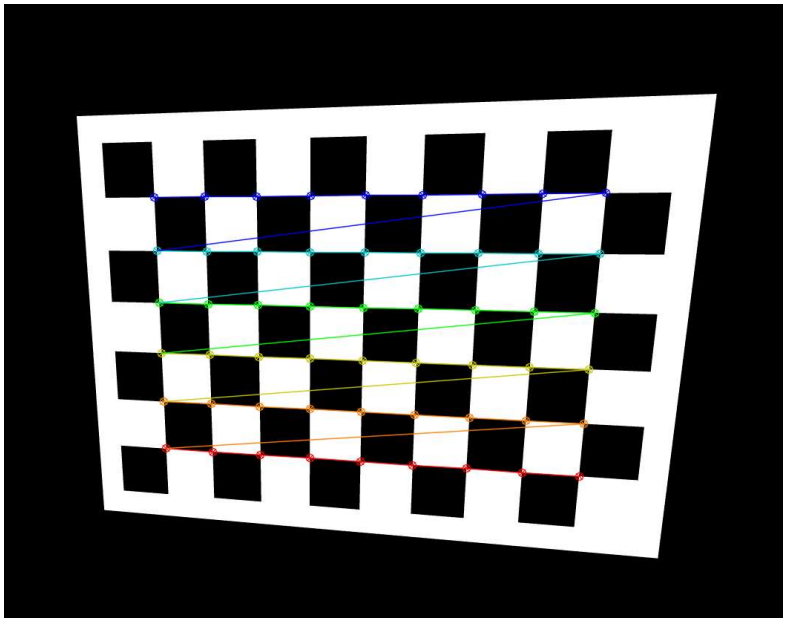
Camera calibration

Checkerboard images with known corner spacing were used to estimate the camera's focal lengths and image centre. These parameters convert image coordinates into measurements used to estimate the marker's distance and orientation.

Calibration setting or result	Value
Internal checkerboard corners	**9 × 6**
Square size	**0.10 m**
Calibration resolution	**1280 × 960**
Training reprojection RMS	**0.101 pixels**
Mean validation reprojection RMS	**0.053 pixels**
Navigation resolution	**2560 × 1920**

The low reprojection error shows that the calibrated model closely matched the detected corner positions.

The calibration parameters are scaled for the higher navigation resolution while retaining the same field of view. Distortion was fixed at zero because the simulated camera uses an ideal pinhole model.



ArUco corner refinement tuning

Corner refinement improves the estimated pixel positions of the marker's corners, affecting the calculated distance and orientation. Nine combinations of window size and relative window parameter were tested at 105 robot poses. Camera measurements were compared with simulator truth.

Window size	Relative parameter	Range RMSE (cm)	Maximum absolute range error (cm)
4	0.3	2.94	8.34
4	0.4	2.57	8.34
4	0.5	2.15	6.40
6	0.3	2.94	8.34
6	0.4	2.55	8.34
6	0.5	2.08	6.40
8	0.3	2.94	8.34
8	0.4	2.55	8.34
8	0.5	2.08	6.40

All settings produced 97 valid estimates from 105 poses, with bearing RMSE approximately 0.013°. Window 8 and relative parameter 0.5 were selected because they produced the lowest range RMSE. Their improvement over window 6 and relative 0.5 was small.

Camera range selection

With the selected settings, range accuracy improved as the robot approached the marker, located near world x = −8.7 m.

Robot world x (m)	Range RMSE (cm)	Valid estimates

0.0	**3.86**	**15/15**
–1.5	**3.07**	**15/15**
–3.0	**1.62**	**15/15**
–4.5	**0.95**	**15/15**
–6.0	**0.32**	**15/15**
–7.5	**0.05**	**13/15**
–8.0	**0.01**	**9/15**

At x = −6 m, the robot was approximately 2.7 m from the marker, with range RMSE of 0.32 cm. This supports the 3 m acceptance limit for the turtle bot to use camera images

Close-range detection failures show that accurate measurements depend on the marker remaining detectable and inside the camera view, angled approaches from close distances did not produce valid estimates. Reported errors include valid estimates only. These tests measured range and bearing errors;

Noise handling

The comparison below retains the original time-based-Q results. The wheel-travel-Q baseline used for controller verification is reported separately at the end of this document.

Independent Gaussian noise was added to the wheel encoder measurements used for posing estimation. The coefficient $0.005m/\sqrt{m}$ gives a cumulative wheel-distance standard deviation of 5 mm over 1 m. This was a chosen test severity, not a measured sensor specification. Physical slip was disabled.

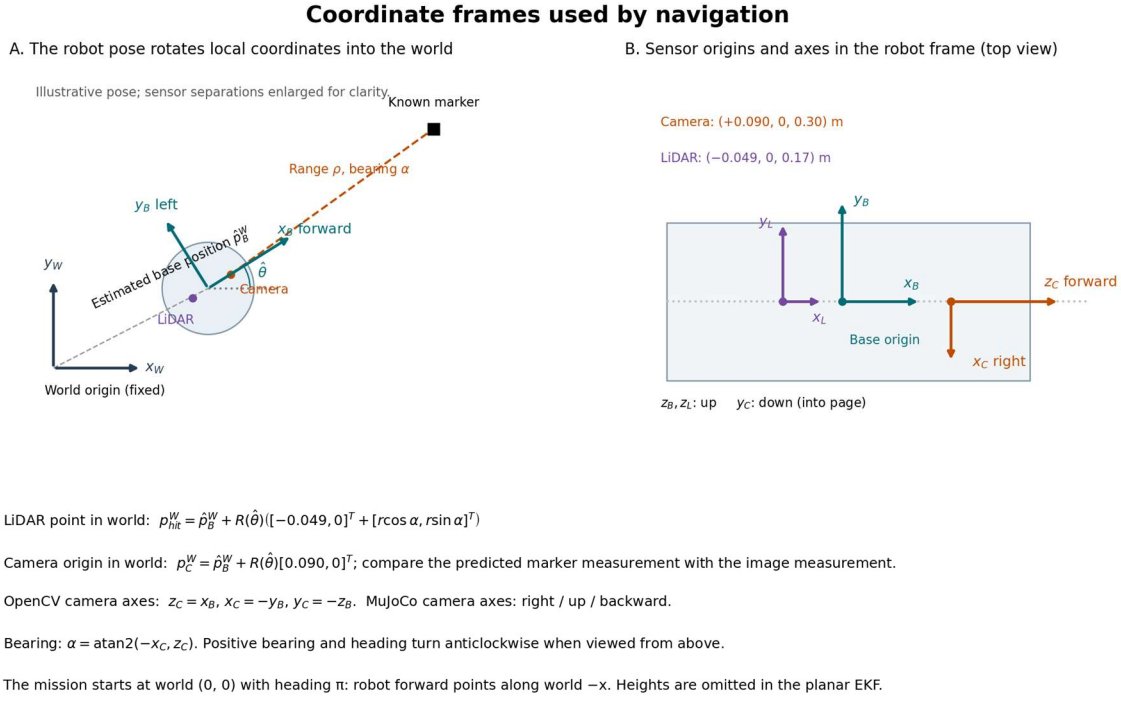
Seed 37 made the noise repeatable in this earlier experiment. The later Q and R comparisons use seeds 37–42 and are reported in the estimation and simulation results sections. EKF uncertainty values are explained in those sections.

Metric	**Wheel-only**	**Fused EKF**
Position RMSE	18.96 cm	8.58 cm
Final localisation error	28.73 cm	0.94 cm
Heading RMSE	2.504°	2.564°

On the same trajectory, camera fusion reduced position RMSE by 55%, although heading RMSE slightly increased. **Docking error was 2.02 cm**. Fusion therefore improved position estimation under noise which allowed it to still meet the docking criteria showing the systems ability to handle noise.

Coordinate frames and transformations

Figure distinguishes the world frame from the robot and sensor frames. The navigation mission begins at world pose (0, 0, π); the earlier odometry validation uses its own start-aligned frame with heading zero. These are different coordinate conventions, not different motion models.



LiDAR point in world: $p_{hit}^W = \hat{p}_B^W + R(\hat{\theta})([-0.049, 0]^T + [r\cos\alpha, r\sin\alpha]^T)$

Camera origin in world: $p_C^W = \hat{p}_B^W + R(\hat{\theta})[0.090, 0]^T$; compare the predicted marker measurement with the image measurement.

OpenCV camera axes: $z_C = x_B$, $x_C = -y_B$, $y_C = -z_B$. MuJoCo camera axes: right / up / backward.

Bearing: $\alpha = \text{atan2}(-x_C, z_C)$. Positive bearing and heading turn anticlockwise when viewed from above.

The mission starts at world (0, 0) with heading π: robot forward points along world −x. Heights are omitted in the planar EKF.

Figure 3. World, base, LiDAR and camera coordinate frames. The left panel is a schematic, not the warehouse layout. Sensor positions in the right panel are body-frame mounting offsets.

The world frame is fixed to the scene. Goals, the known marker and the EKF position are expressed in it. The base frame moves with the robot: x points forward, y left and z up. At the declared initial heading of π, robot forward points along world negative x. The earlier standalone odometry validation used its own start-relative frame, which is distinct from this mission world frame.

The LiDAR is mounted at (−0.049, 0, 0.17) m relative to the base, with aligned axes. A measured beam is converted to a horizontal Cartesian point, translated by the mounting offset, and rotated and translated into the world using the estimated pose. A hit 1 m directly forward is therefore (0.951, 0) m relative to the base, or approximately (−0.951, 0) m at the declared launch pose.

The camera is mounted at (0.09, 0, 0.30) m. OpenCV uses right, down and forward axes; MuJoCo uses right, up and backward, requiring a sign reversal of the latter two coordinates. The planar EKF accounts for the horizontal camera offset and assumes negligible pitch and roll. It compares image-derived range, bearing and marker-derived heading against measurements predicted from the estimated pose and known marker. Simulator pose is not supplied to this calculation.

Estimation and control

EKF prediction uncertainty

The EKF predicts position and heading from the wheel encoders and corrects that prediction when the camera observes the charging-station marker. We compared two ways of calculating Q, the uncertainty added at each prediction. The original model adds uncertainty in proportion to elapsed time. The revised model uses the measured travel of each wheel. For each wheel, travel-error variance is the absolute wheel distance multiplied by 0.005^2, matching the noise injected in the simulation. Equal wheel errors affect forward distance, while the difference between the two errors affects heading.

The estimated state is robot position and heading, [x, y, θ]. Each prediction uses measured left and right wheel rotation in the circular-arc motion model. The state Jacobian F propagates covariance using $P^- = FPF^T + Q$. Camera observations contain horizontal range, bearing and marker-derived world heading. The observation model includes the 0.09 m forward camera offset. Angular residuals are wrapped to ±π. Accepted observations update the pose and covariance using the Kalman gain and Joseph covariance update. The controller receives the estimated pose; simulator pose is used for evaluation.

For wheel-travel Q, each wheel has travel-error variance $k^2|\Delta s|$, where $k = 0.005\ \text{m}/\sqrt{\text{m}}$. A small-step wheel-to-pose Jacobian G maps these variances into $Q = G\,\text{diag}(k^2|\Delta sL|, k^2|\Delta sR|)\,G^T$. The implementation uses measured travel and the midpoint heading as approximations. This models the deliberately injected encoder errors; additional slip and wheel-model errors remain possible. The mission starts with standard deviations of 1 cm in x and y and 1° in heading.

EKF matrices and local observability

For the circular-arc prediction, let δx and δy be the predicted displacement already defined in the motion model. The state Jacobian is $F = [[1, 0, -\delta y], [0, 1, \delta x], [0, 0, 1]]$. Let $\psi = \hat{\theta} + \Delta\theta/2$. The implemented small-step wheel Jacobian is $G = [[\cos\psi/2, \cos\psi/2], [\sin\psi/2, \sin\psi/2], [-1/b, 1/b]]$. Then $Q = G\,\text{diag}(k^2|\Delta s_L|, k^2|\Delta s_R|)\,G^T$, with b = 0.288 m and $k = 0.005\ \text{m}/\sqrt{\text{m}}$. This is a wheel-travel noise approximation, not an exact slip model.

Let the known marker position be m = (−8.697, 0) m and forward camera offset a = 0.09 m. The camera centre is $c = [x + a\cos\theta, y + a\sin\theta]^T$. Define $d_x = m_x - c_x$, $d_y = m_y - c_y$, $q = d_x^2 + d_y^2$ and $\rho = \sqrt{q}$. The predicted measurement is $h(x) = [\rho, \text{wrap}(\text{atan2}(d_y, d_x) - \theta), \text{wrap}(\theta)]^T$. The third measurement is world heading reconstructed from the observed marker normal and the known marker orientation.

The observation Jacobian has rows:

$H_1 = [-d_x/\rho, -d_y/\rho, a(d_x \sin\theta - d_y \cos\theta)/\rho]$

$H_2 = [d_y/q, -d_x/q, -1 - a(d_x \cos\theta + d_y \sin\theta)/q]$

$H_3 = [0, 0, 1]$.

For ρ > 0, det(H) = 1/ρ, so H has rank 3. An accepted range–bearing–heading observation locally distinguishes all three planar pose components. This is a local, instantaneous measurement-rank check; marker visibility, pose ambiguity, calibration bias and correlated measurement errors still limit practical accuracy. Without the camera, wheel increments propagate a pose from its initial condition but provide no independent absolute-position correction.

The update uses $v = z - h(\hat{x}^-)$, with angular residuals wrapped, $S = H P^- H^T + R$, $K = P^- H^T S^{-1}$ and $\hat{x}^+ = \hat{x}^- + Kv$. The code rejects nonfinite innovations or $v^T S^{-1} v > 25$. It uses the Joseph update $P^+ = (I - KH)P^-(I - KH)^T + KRK^T$ and symmetrises P. This broad innovation threshold is a fixed diagnostic gate, not a calibrated probability guarantee. The chosen R and its empirical comparison are reported below.

LQR design and actuator implementation

The controller minimises the infinite-horizon discrete cost $J = \Sigma(e^T Q_LQR\ e + \omega^2 R_LQR)$. $Q_LQR = diag(1/0.10^2, 1/(15\pi/180)^2) = diag(100, 14.5903)$, and $R_LQR = 1/0.5^2 = 4$. These weights normalise 10 cm lateral error, 15° heading error and 0.5 rad/s yaw-rate effort. They are engineering tuning scales, not hard tolerances or a proven optimum for all missions. The EKF covariance Q and R are separate quantities from these controller weights.

The discrete algebraic Riccati equation is $S_L = Q_LQR + A^T S_LA - A^T S_LB(R_LQR + B^T S_LB)^{-1} B^T S_LA$. The implementation iterates to a maximum element change below 10^{-10}, with a 20,000-iteration limit. It forms $K_L = (R_LQR + B^T S_LB)^{-1} B^T S_LA$ and commands $\omega = -K_L\ e$. Both errors are calculated from the EKF pose.

The six scheduled gains [K_y, K_θ], rounded, are: v = 0.025 m/s: [4.7593, 1.8822]; 0.050: [4.7519, 1.9416]; 0.075: [4.7447, 1.9991]; 0.100: [4.7377, 2.0549]; 0.130: [4.7296, 2.1198]; 0.160: [4.7218, 2.1826]. K_y has units rad/(m·s), and K_θ has units s^{-1}. The nearest speed selects the gain; the code does not interpolate. The nominal closed-loop eigenvalues of A − BK lie inside the unit circle at each design speed. This is a local, unsaturated-model result, supported separately by the recovery tests.

Forward speed is clip(0.6 × remaining along-segment distance, 0.025, 0.16) m/s. TRACK limits yaw rate to ±0.70 rad/s. ALIGN holds v = 0 and applies ω = clip(−1.5e_θ, −0.70, 0.70) until $|e_\theta| \le 4°$. A tracking heading error above 50° returns to ALIGN. Intermediate segment endpoints use 2 cm estimated distance; delivery bays use 3 cm. An along-track overshoot greater than 6 cm stops tracking and requests replanning. These thresholds are controller logic, not measurements of true arrival accuracy.

Body commands are converted to wheel rates: $\omega_L = (v - b\omega/2)/r$ and $\omega_R = (v + b\omega/2)/r$, with r = 0.033 m. If either target exceeds 5.5 rad/s, both are scaled by the same factor, preserving their ratio. A PI trim around the MuJoCo velocity actuators uses encoder-rate error, proportional gain 0.5 and integral gain 2 s^{-1}. The integral contribution is bounded to ±3 rad/s with conditional anti-windup; final actuator velocity commands are clipped to ±7.88 rad/s. Stop commands reset the integral.

The simulation physics step is 0.002 s, navigation control period is 0.05 s and camera observation period is 0.5 s. The wheel servo uses measured wheel rates; the deliberately added noise affects localisation increments, rather than the inner servo feedback. Simulator position and heading do not feed the path tracker, planner transforms or docking controller. Simulator geometry is used to generate sensor readings and to evaluate results.

Mission switching and visual docking

LiDAR scans and estimated pose populate the planner grid; the planner does not receive a prefilled obstacle map. The navigator plans a route to each predefined goal, aligns to a segment and tracks it with LQR. Live scan clearance can stop commanded motion independently of the route tracker. After the three bays and charger approach, the system changes to low-speed visual docking. Docking uses polar pose feedback rather than the straight-segment LQR.

For target displacement $\delta = p_d - \hat{p}$, define $d = \|\delta\|$, $\alpha = \text{wrap}(\text{atan2}(\delta_y, \delta_x) - \hat{\theta})$ and $\beta = \text{wrap}(\theta_d - \hat{\theta} - \alpha)$. Docking commands $v = \min(0.06, 0.5d)$ m/s when $|\alpha| < 35°$, otherwise $v = 0$; $\omega = \text{clip}(1.4\alpha - 0.5\beta, -0.30, 0.30)$ rad/s. Within 2 cm estimated distance, translation stops and heading verification uses $\omega = \text{clip}(-1.5\,\text{wrap}(\hat{\theta} - \theta_d), -0.25, 0.25)$.

Accepted camera observations must be no more than 1.2 s old; otherwise WAIT CAMERA commands zero motion. Confirmation requires the independently reconstructed visual base pose to be within 5 cm and 3° of the target and the fused pose within 3 cm and 3°. Three consecutive qualifying observations are required. A failed observation/qualification or stale-camera condition resets the count. The dock phase has a 60 s timeout; successful docking is followed by a two-second hold before results are finalised.

The final target is $(-8.197, 0, \pi)$, 0.50 m from the marker plane measured from the base. This is a stand-off target, not physical charger contact. Success is declared using sensed and estimated poses. The final true-pose errors are reported separately to check the 3 cm and 3° evaluation objective.

Simulation results

Effect of wheel-travel Q under encoder noise

We ran each Q model with seeds 37 - 42, using the same encoder-noise strength, normal floor traction, camera settings and LQR gains. Each seed selects a repeatable sequence of encoder errors. Goal 2 provides a direct check of prediction uncertainty because all five completed pairs reached it before the first accepted camera update. Both filters therefore produced the same estimated heading within each pair.

Measure	**Time-based Q**	**Wheel-travel Q**
Goal 2 heading error, RMS across five seeds	3.47°	3.47°
Goal 2 predicted heading uncertainty	1.89°	3.81°
Mean position RMSE over five completed missions	8.80 cm	8.76 cm
Camera-confirmed docks across six attempted seeds	5/6	5/6

For each run, we took the square root of the EKF heading variance at Goal 2, converted it to degrees, then calculated the RMS across seeds 37, 38, 39, 40 and 42.

We selected wheel-travel Q because its predicted heading uncertainty was closer to the observed error across these trials. Position RMSE differed by only 0.04 cm, and both models docked on the same five seeds. The

evidence supports a better estimate of wheel-driven uncertainty, with little change in average localisation accuracy.

Seed 41 Planning Stall

The terminal outputs for both Q types for seed 41 showed the TurtleBots successfully arrive at Goals 1 and 2. Shortly after leaving goal 2 and attempting to navigate to goal 3, the planner repeatedly reported “NO FOOTPRINT CLEARANCE” and returned zero route points while attempting Goal 3. With no accepted camera update before the stall, both filters followed the same wheel-based estimate. Seed 41 counts as an incomplete mission for each model. Only terminal transcripts were available, so its heading uncertainty and trajectory RMSE are excluded from the numerical comparison. The transcripts establish the planner’s reason for commanding a stop but do not identify which map cells prevented a route.

Camera uncertainty measurement and R selection

The camera accuracy scripts compared image-derived range, bearing and world heading with simulator truth at different static positions and viewing angles. The test and validation sets used separate viewpoints. Measurements passed the same front-facing and 2-pixel reprojection checks as the mission, followed by the measured-range limit of 3 m. There were 37 usable test samples and 23 usable validation samples within that limit. Error covariance was calculated in metres and radians. Mean error was recorded separately to expose systematic offsets.

Quantity	Test SD	Validation SD	Original R SD	Candidate R SD
Range	0.856 mm	0.291 mm	20 mm	5 mm
Bearing	0.00293°	0.00240°	0.5°	0.05°
World heading	0.229°	0.083°	5°	1°

The mean bearing error was −0.01255° in the test set and −0.01181° in validation. Its repeatability indicates a small systematic offset. The measured covariance describes variation across static viewpoints and excludes that mean bias. The validation samples within the range gate were concentrated around 1.86 m and 0.86 m, leaving the approximately 0.57 m docking distance untested by that set. Motion, pitch and roll, and repeated correlated camera observations can also differ from this static test.

We therefore tested a diagonal R with standard deviations of 5 mm, 0.05° and 1°. These were rounded tuning choices with margins above the static errors. Both the original and candidate R omit correlations between measurements derived from the same marker corners. Lower R increases camera influence and also tightens the innovation gate. The mission rejects updates when the normalised innovation squared exceeds 25; this threshold was held fixed.

Mission comparison of camera uncertainty settings

Both R settings were evaluated using wheel-travel Q, encoder noise k = 0.005 m/√m, normal traction and the same camera and LQR configuration. The paired seeds were 37–42. Seed 41 stalled before camera corrections under both settings, as described above. The table reports position RMSE for the five completed pairs. Each value summarises that run’s actual trajectory; different corrections can change subsequent steering and observations.

Seed	Original R position RMSE	Candidate R position RMSE
37	8.59 cm	8.08 cm
38	3.99 cm	4.63 cm
39	15.48 cm	15.43 cm
40	10.01 cm	10.10 cm
42	5.73 cm	5.90 cm
Mean	8.76 cm	8.83 cm

Both settings completed five of the six attempted seeds with camera-confirmed docking. All recorded in-range camera corrections were accepted in the five candidate-R runs. The candidate improved seed 37, reducing actual Goal 3 distance from 10.55 cm to 5.63 cm, but seed 38 worsened from 3.46 cm to 8.32 cm. Bay distances are reported as accuracy measurements. Arrival is declared using a 3 cm estimated-position threshold, and actual bay error has no pass/fail cutoff.

Selected settings and limitations

The selected configuration combines wheel-travel Q with the original camera R. Wheel-travel Q gave a predicted heading spread closer to the observed errors before camera correction. The original R produced slightly lower mean position RMSE across the five completed pairs. The 0.07 cm difference is small and does not establish statistical superiority; the candidate provided no consistent improvement sufficient to justify replacing the original setting.

The selected camera covariance is $R = \mathrm{diag}(0.02^2, (0.5\pi/180)^2, (5\pi/180)^2)$, ordered as range, bearing and heading. Its diagonal values are approximately $[4.00 \times 10^{-4}, 7.62 \times 10^{-5}, 7.62 \times 10^{-3}]$ in m^2, rad^2 and rad^2. These are conservative mission tuning values supported by the comparison. The smaller covariance measured in the static tests is retained as sensor-characterisation evidence.

The results show that uncertainty tuning affects both confidence and subsequent camera corrections. During camera loss, the estimate still accumulates wheel errors. Seed 41 remains an integration limitation requiring investigation of the LiDAR planner and its map. No numerical seed 41 trajectory was available, so its transcript contributes only to the completion count and failure description. The five completed pairs provide preliminary empirical support for the chosen settings, with limited coverage of conditions and no formal full-state consistency test.

Wheel response verification

The wheel-response tests checked whether the simulated robot achieved the speeds requested by the controller. The current WheelServo was used with its existing proportional gain of 0.5, integral gain of 2 s^{-1} and actuator-command limit of ±7.88 rad/s. Each of seven cases reset to the declared XML starting pose, settled for one second, drove for 2.5 or 3 seconds, and then commanded a stop for two seconds. The slip plate had normal floor traction.

The servo receives ideal wheel-encoder rates in the mission, so this separate inner-loop test used the same measurements. It did not run the EKF or LQR. Encoder noise remained enabled in the full-mission test. Simulator position and heading were used only to score actual motion. All seven wheel-response cases completed.

Actual speeds in the table are averages over the final second of each driving phase. The response time is the first sampled time at which both wheel rates entered ±5% of their respective nonzero targets and remained inside that band for the rest of the driving phase. This is a response metric, not a project success threshold.

Case	Requested v (m/s)	Actual v (m/s)	Requested ω (rad/s)	Actual ω (rad/s)	5% response (s)
Slow straight	0.025	0.02435	0	0.00001	2.04
Medium straight	0.100	0.09937	0	−0.00010	0.85
Cruise straight	0.160	0.15939	0	−0.00012	0.30
Left rotation	0	−0.00025	+0.500	+0.49033	1.26
Right rotation	0	+0.00019	−0.500	−0.49065	1.26
Left curve	0.100	0.09928	+0.300	+0.29926	1.31
Right curve	0.100	0.09964	−0.300	−0.29884	1.31

The straight-speed shortfall was 2.59% at 0.025 m/s, 0.63% at 0.10 m/s and 0.38% at 0.16 m/s. Rotation-rate shortfall was approximately 1.9% in both directions. No actuator commands reached their limits in these tests, and sampled wheel rates approached their targets without exceeding them. After 0.5 seconds of stopping, the largest recorded absolute wheel rate was approximately 0.00027 rad/s.

These results support retaining the existing servo and wheel geometry for the next controller tests. They do not establish calibration over all loads or floor conditions. Low-speed response took about two seconds, so the tracker's assumption that requested motion is achieved immediately is only an approximation. This delay should be considered when assessing short segments and transitions.

Baseline path tracking with encoder noise

The baseline used wheel-travel Q, the original camera standard deviations of 0.02 m, 0.5° and 5°, encoder-noise coefficient 0.005 m/Vm, seed 37 and normal floor traction. LQR weights, scheduled gains, speed limits and planning logic were unchanged. The trajectory, goal and camera-correction CSV data matched the preceding seed 37 run exactly, confirming that the diagnostic additions did not change this run's motion.

The table covers 2,834 active TRACK samples without a stop override. Cross-track error is the perpendicular distance from the active straight segment; heading error is relative to its direction. Estimated and actual errors use the same reference. Alignment, planning and docking are excluded.

Tracking measure	Estimated pose	Actual pose for evaluation
Cross-track RMS	0.25 cm	6.54 cm
Maximum absolute cross-track error	1.09 cm	18.25 cm
Heading RMS	0.72°	2.68°
Maximum absolute heading error	3.98°	8.81°

The controller followed its perceived route closely. The larger actual error is consistent with accumulated localisation error: stronger gains cannot directly correct an error the estimator does not report. The reference route was also generated using the estimated pose, so it is not an independent ideal path.

No LQR steering request reached ±0.70 rad/s. Wheel targets were scaled to their 5.5 rad/s limit in seven samples. Actuator commands reached ±7.88 rad/s in 29 of 3,722 control samples, all immediately after ALIGN changed to TRACK. This rotation-to-forward transition warrants investigation, but brief limiting alone does not demonstrate instability. No overshoot-replanning event was recorded.

Arrival	Estimated distance (cm)	Actual distance (cm)	Localisation error (cm)
Goal 1	2.89	4.47	1.70
Goal 2	2.89	12.65	11.36
Goal 3	2.97	10.55	9.89
Charger approach	2.92	2.33	0.64
Stand-off dock	1.90	1.80	0.16

All three bay arrivals were declared using the 3 cm estimated-position threshold, followed by camera-confirmed stand-off docking. Actual bay distances remain accuracy measures without a pass/fail cutoff. Whole-mission localisation RMSE was 8.59 cm, which measures estimated-versus-actual position and is distinct from cross-track RMS. No collision-contact count was logged, so completion does not by itself verify the zero-contact objective.

Further controller validation

The earlier 5 cm sideways-recovery example in the state-space section is retained as a separate test with noise disabled. These tests extend that evidence using encoder noise and the current estimator and controller settings. Gains and wheel-servo settings were fixed while recovery, alignment and corner behaviour were measured.

Case	Initial condition	What it checks
On-path reference	Zero sideways and heading offset	Reference run for the same fixed straight segment
Sideways recovery	±5 cm from the line; parallel heading	LQR recovery in both lateral directions
Heading recovery	On the line; heading ±10°	Small-angle LQR recovery, starting in TRACK
Alignment transition	On the line; heading +30°	Existing ALIGN rotation followed by forward TRACK
Corner	Two fixed straight segments at 90°	Segment handover, alignment and subsequent tracking

Each case uses a declared test starting pose to place the simulated robot and initialise the estimator. This represents a known physical offset from the path, not a deliberately wrong pose estimate. The initial estimate is never copied from simulator truth. The test fixture is reset before each trial; no pose resets occur during a trial. All feedback uses the estimated state. Simulator truth is used only for scoring.

The supplied test script uses fixed reference segments in the open corridor, bypassing route planning so that replanning cannot hide tracking error. LiDAR stopping remains active, and camera measurements retain the mission's detection, range and innovation gates. These short routes are expected to lie outside the 3 m camera correction range; the script records the actual accepted-update count rather than assuming it is zero.

The seven cases were run with seeds 37, 39 and 42. Estimated and actual tracking RMS, endpoint error, recovery time, alignment duration and command limiting were compared. A diagnostic recovery band of ±1 cm cross-track and ±2° heading sustained for one second is recorded separately from endpoint arrival. It is not a new mission pass/fail requirement. The corner is assessed by segment and transition.

A gradual increase in forward speed after alignment was identified as a candidate change. Any trial should be compared against the unchanged controller before changing LQR weights. W4 evaluates the completed recovery tests; no gain or speed-ramp improvement is claimed.

LQR recovery results

Every trial reached its estimated endpoint. No LiDAR-stop or segment-overshoot status was recorded, and no LQR steering request reached its angular-rate limit. Configuration files and source hashes matched across the three seeds apart from the seed value. Encoder noise, wheel-travel Q and the original camera R remained enabled as specified. No camera correction was accepted in any trial, so localisation relied on wheel prediction throughout.

The table reports first entry into the joint ±1 cm cross-track and ±2° heading band, sustained for one second of active tracking. Times are measured from the beginning of each case. A dash means this interval was not observed. Entering the band once does not establish that the robot remained there until arrival. Endpoint distances are actual distances at estimated arrival; they have no pass/fail cutoff.

Case	Actual band-entry time (s) Seed 37 / 39 / 42	Actual endpoint distance Range across seeds (cm)
On-path reference	0.00 / 0.00 / 0.00	1.64 - 2.82
Sideways +5 cm	5.79 / - / 4.96	1.69 - 2.67
Sideways −5 cm	- / - / 6.26	1.76 - 2.93
Heading +10°	0.75 / 0.75 / 0.75	1.66 - 2.53
Heading −10°	0.75 / - / 0.75	1.71 - 3.00
Alignment +30°	1.99 / 1.94 / 1.94	1.51 - 2.13
Corner, second segment	12.28 / 12.13 / 12.23	0.85 - 1.56

Sideways recovery: the estimate entered the joint band in all six trials, after 4.49 - 5.22 s. Actual motion met the same sustained band in three of six trials. The initial 5 cm actual offset reduced to 0.01 - 2.25 cm in absolute value at the final active tracking sample. The two measures therefore show useful correction with residual drift, rather than guaranteed centimetre-level accuracy.

Heading recovery: all six estimates entered the band, while actual motion did so in five of six trials. The final active-tracking heading error was within 0.89° in every heading-offset case. Seed 39 with a -10° initial offset did not sustain the joint band because lateral error remained too large. Both angular and lateral response must be checked.

Alignment and corner: the +30° cases entered TRACK at 1.68 - 1.73 s. Corner realignment lasted 3.10 - 3.15 s between the segment handover and resumed tracking. All three second corner segments entered the actual joint band, 0.20 - 0.25 s after tracking resumed. Their actual cross-track RMS was 0.27 - 0.62 cm. The endpoint-distance range across all 21 trials was 0.85 - 3.00 cm.

Recovery traces and interpretation

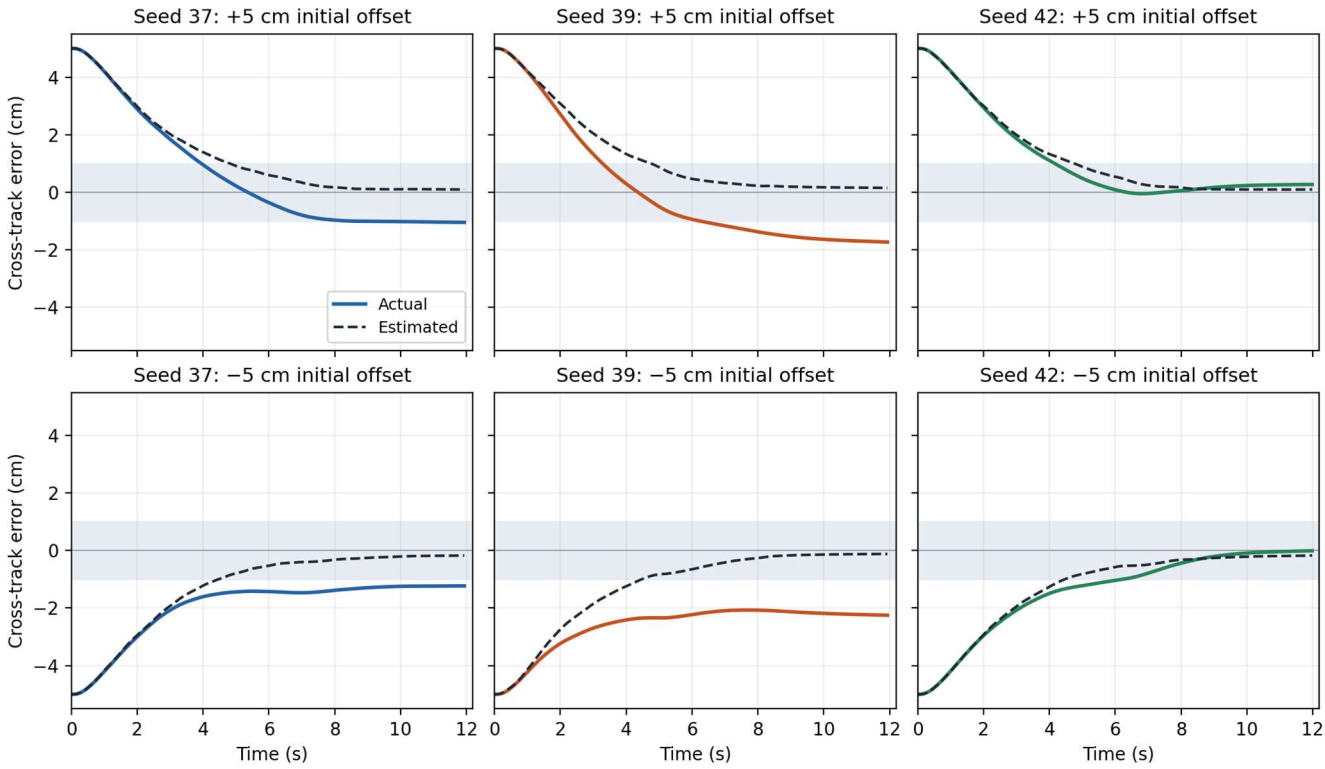


Figure 2. Sideways-offset tests with encoder noise. Solid curves show actual error; dashed curves show estimated error. The shaded region is the ±1 cm cross-track band only; the recovery metric additionally requires heading within ±2° for one second. Plots stop at the final active tracking sample.

The estimated cross-track error reduced towards zero in both directions. Actual curves depended more strongly on seed: seed 39 developed a similar negative offset even in the on-path reference, which ended active tracking at -2.01 cm. This supports wheel-only localisation drift as a contributor to the residual errors. It does not prove that all physical tracking error is caused by estimation, and the band-entry times must not be interpreted as permanent settling times.

The forward-speed request jumped from zero to 0.16 m/s when ALIGN changed to TRACK. Each alignment and corner trial hit the ±7.88 rad/s actuator-command limit for two consecutive samples immediately after this change, approximately 0.10 s. All continued tracking and reached their estimated endpoints. Brief saturation is a measured transition effect, not evidence by itself that the controller is unstable.

The current controller demonstrated recovery and corner handling under the tested conditions. Retain the present gains as the verified baseline. A forward-speed ramp can be evaluated as an optional iteration to reduce transition saturation, using the same cases and seeds and checking that recovery and endpoint errors do not worsen. The tests do not resolve the seed 41 full-mission planner stall, because this fixture bypasses the planner, and no collision-contact count was recorded.

Collision monitoring verification

The instrumented seed 37 mission reached all three bays and completed camera-confirmed stand-off docking. Across 94,681 physics samples spanning 189.362 s, the monitor recorded zero obstacle-contact episodes, zero contact samples and zero sampled contact duration. This verifies the zero-obstacle-contact objective for this particular completed run under the simulated collision model.

The monitor checks contacts between robot geometry and external geometry after every physics step. Named floor sections and the slip plate are excluded as support surfaces, along with robot self-contact. A contact qualifies if its solver constraint is active or its signed geometry separation is zero or negative; positive-margin active contacts are therefore counted conservatively. A sustained sequence counts as one episode rather than one collision per physics step. The counter is evaluation only and does not alter navigation, control or estimation.

Monitoring starts after the initial settling period. The result does not establish collision-free operation for every seed or floor condition, and it cannot assign counts retrospectively to previous runs. The quantitative result above comes from the saved mission contact files.

Run interruptions and measurement loss

The instrumented seed 41 rerun stalled during Goal 3 planning. Ctrl+C ended the run at approximately 89.574 s with the outcome “Interrupted by user; partial results saved”. The output retained 1,755 trajectory samples, 1,755 controller samples, goal records, controller events and the LiDAR map. Goals 1 and 2 had been declared reached; Goal 3 and docking were not reached. This verifies handled program interruption and partial-result preservation while the robot was already stationary. It does not demonstrate restart/resume or emergency stopping from motion.

A separate earlier EKF integration test disabled camera corrections from 3 to 5 s. Ten updates were skipped, wheel prediction continued and camera corrections resumed. That test used an earlier configuration without deliberately added encoder noise. It is evidence of temporary measurement-loss handling, distinct from the Ctrl+C test and from the natural periods without marker visibility in the noisy missions.

Seed 41 planner diagnosis

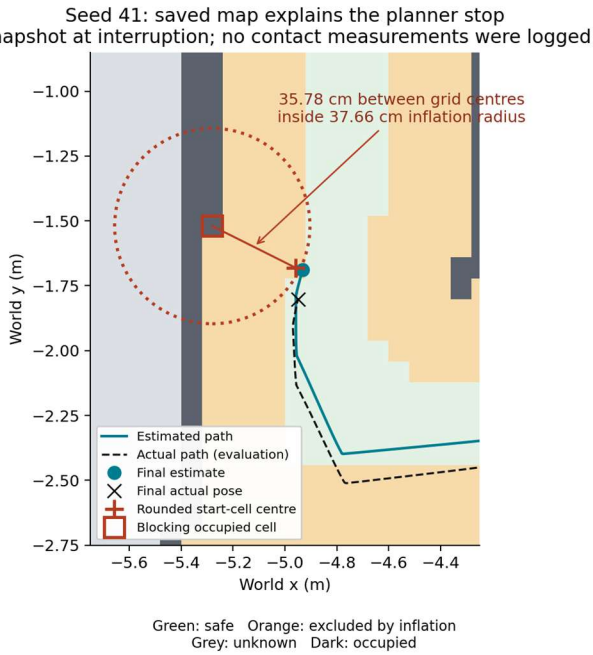


Figure 4. Final saved map at interruption. Occupied-cell inflation excludes the rounded robot start cell. This is not a map snapshot at the first failure.

At 75.324 s, the remaining route became invalid and replanning returned NO FOOTPRINT CLEARANCE. The controller entered PLAN and commanded zero motion. In the final map the rounded start cell at (−4.96, −1.68) m is observed free, with log odds −4, but marked unsafe. An occupied cell at (−5.28, −1.52) m lies 0.35777 m away, within the 0.37657 m obstacle-inflation radius. No unknown cell lies within the corresponding 0.31 m unknown-space inflation radius. Occupied-cell inflation therefore explains the continuing planning refusal.

The live scan reported approximately 0.130 m stationary clearance. The scan and accumulated world-frame grid use different geometry and discretisation, so their decisions can differ. No camera correction had been accepted, and final localisation error was 11.56 cm. Pose drift and grid rounding are plausible contributors, but the final map cannot identify which historical observation created the blocking cell. The immediate stop mechanism is established; its full mapping history remains unresolved.

Position RMSE was 7.40 cm and heading RMSE 2.187° over this incomplete run. These values are not directly comparable with completed-course RMSE and are not added to the earlier paired complete-run averages. Seed 41 remains an incomplete mission. Its rerun preceded contact monitoring, so its collision count remains unmeasured.

Quantitative evaluation and success definitions

This update adds the final three demo trials, consolidates the earlier repeated-trial evidence and diagnoses failures and trade-offs. Only sections labelled NEW N1 to NEW N4 identify changes in this revision. Previous

review labels and change records have been removed; historical experiment configurations remain distinguished because they are not interchangeable.

A completed mission requires all three estimated bay arrivals followed by camera-confirmed stand-off docking. Bay arrival uses a 3 cm estimated-position threshold; actual bay distance is an accuracy measurement, not an additional pass/fail test. Localisation position RMSE measures estimated-versus-true planar position over logged control samples. Tracking RMS measures distance from the active reference segment and is a different quantity. Final dock errors below use the final pose report, whereas goal-arrival tables sample the first declared arrival. Simulator truth is used only for evaluation.

Experiment	Observed completion	Meaning and limitation
Earlier Q comparison	5/6 for each Q model (83.3%)	Seeds 37–42; seed 41 incomplete in both configurations.
Earlier R comparison	5/6 for each R setting (83.3%)	Same seed set; overlapping evidence, not another independent 12-trial cohort.
Fixed-route recovery	21/21 estimated endpoints (100%)	Seven cases × three seeds; bypasses full-mission planner.
Final demo conditions	3/3 missions (100% observed)	One trial per condition, seed 37; no per-condition reliability estimate.

Repeated trials establish sensitivity to random encoder errors, while the three demo conditions add physical traction variation. These experiments must not be pooled into one overall success rate: configurations, paths, endpoints and failure criteria differ. Completed-run averages exclude the incomplete seed 41 trajectory but its failure remains in the completion denominator. The demo video is complete as reported by the operator; the numerical assessment here is based on its saved run files.

Final demo results and interpretation

The final demonstration video can be found at https://youtu.be/mWIeYK-iH-M?si=dDp2DcB_gHl5Da40

The final demonstration video can be found at https://youtu.be/mWIeYK-iH-M?si=dDp2DcB_gHl5Da40All three cases used wheel-travel Q, the original camera R, the existing LQR and wheel servo, and seed 37. Mode 1 disables deliberately added encoder noise; it does not eliminate camera, dynamics or modelling error. Mode 2 adds encoder noise k = 0.005 m/√m with normal traction. Mode 3 retains that noise and reduces insert sliding friction from 1.0 to 0.04. The physical insert remains present in every case. The desired base pose is (−8.197, 0, π), 0.50 m from the marker. This is stand-off docking, not electrical charging contact.

Mode	Duration s	Position RMSE cm	Heading RMSE deg	Final dock XY cm	Final dock yaw deg
1	184.36	1.06	0.329	1.90	1.171
2	191.86	8.53	2.566	1.88	0.315
3	191.36	8.45	2.460	1.99	1.943

Duration is the monitored mission interval, including the final confirmation/hold period and excluding initial settling. All three status files report ALL GOALS REACHED + CAMERA-CONFIRMED STAND-OFF DOCK. All three final true dock poses satisfy the stated 3 cm and 3° evaluation objective. Zero obstacle contacts were recorded over 92,181, 95,931 and 95,681 physics samples respectively (283,793 total). Support surfaces and self-contact are excluded; this result is limited to these runs and their collision geometry.

Mode	Actual Goal 1 cm	Actual Goal 2 cm	Actual Goal 3 cm
1	3.35	3.78	3.82
2	4.47	12.65	10.55
3	4.47	12.65	10.58

Added encoder noise increases position RMSE from 1.06 to 8.53 cm and raises Goal 2 distance from 3.78 to 12.65 cm. The estimated arrival remains near 3 cm, so the larger true error is primarily a localisation limitation rather than failure to approach the estimated goal. Camera corrections near the charger subsequently reduce the final localisation errors to 0.04, 0.08 and 0.66 cm. Accurate final docking therefore does not imply equally accurate delivery at every earlier bay.

Mode	Wheel-only RMSE cm	Fused RMSE cm	Position RMSE reduction	Final localisation cm
1	2.03	1.06	47.8%	0.04
2	19.07	8.53	55.2%	0.08
3	17.72	8.45	52.3%	0.66

The wheel-only estimator is a shadow using the same noisy wheel increments on the EKF-controlled trajectory. It is not a separate odometry-controlled mission. Fusion improves position RMSE in all three cases. Heading RMSE changes from 0.320° to 0.329° in mode 1 and from 2.486° to 2.566° in mode 2, but improves from 3.374° to 2.460° in mode 3. Thus better position accuracy does not guarantee better whole-run heading accuracy. Of the accepted camera updates, 48/51, 45/54 and 45/54 immediately improved position; individual corrections can worsen error even when overall fusion is beneficial.

The slip disturbance was physically exercised: mode 3 recorded 15.54 s of tyre–plate contact at friction 0.04. Left/right accumulated absolute rolling-distance mismatches were 6.93/4.73 cm, or 4.31/3.21% of wheel travel, versus 0.39/0.38 cm and approximately 0.26/0.26% with noise and normal traction. Twenty-one camera updates were accepted during plate contact in each noisy run. Mode 3 position RMSE was slightly lower than mode 2 (8.45 versus 8.53 cm), despite greater rolling mismatch. This does not show that slip improves performance: camera correction timing, trajectory and error cancellation affect aggregate RMSE. The physical mismatch and larger final localisation error expose the disturbance more directly.

Repeated trials and quantified trade-offs

For the earlier five completed Q pairs, observed Goal 2 heading RMS was 3.47°. Predicted heading uncertainty was 1.89° with time-based Q and 3.81° with wheel-travel Q. The uncertainty-to-error ratios were therefore approximately 0.54 and 1.10. Wheel-travel Q better represented the observed spread, while mean position RMSE changed by only 0.04 cm (8.80 to 8.76 cm) and completion stayed 5/6. This is evidence about heading uncertainty at one checkpoint, not a full-state NEES consistency test.

Reducing camera standard deviations from 20 mm/0.5°/5° to 5 mm/0.05°/1° increased camera weighting and tightened the fixed innovation gate. Across five completed pairs, mean position RMSE rose from 8.76 to 8.83 cm. Seed 37 improved by 0.51 cm, but seed 38 worsened by 0.64 cm; its Goal 3 distance increased from 3.46 to 8.32 cm while seed 37 improved from 10.55 to 5.63 cm. Both settings completed 5/6 attempted missions. The candidate therefore did not provide a consistent accuracy benefit. The small mean difference does not establish statistical superiority, and settling-time responsiveness was not measured.

The baseline tracker achieved estimated cross-track RMS of 0.25 cm versus actual RMS of 6.54 cm, approximately 26 times larger. Increasing steering gains cannot directly correct pose errors absent from feedback. In the 21 recovery trials, all estimated endpoints were reached, but actual sustained recovery occurred in only 3/6 sideways trials (50%) and 5/6 heading trials (83.3%). All three alignment trials and all three second corner segments met the joint recovery band. Actual endpoint distances ranged from 0.85 to 3.00 cm. The diagnostic band requires ±1 cm lateral and ±2° heading error for one second; entering it does not guarantee permanent settling.

Control effort was bounded but transitions briefly saturated: 29/3,722 baseline actuator-command samples (0.78%) reached the limit, with no LQR angular-rate saturation. Alignment/corner trials each had approximately 0.10 s of actuator limiting after the forward-speed step to 0.16 m/s. Straight-drive response improved from 2.04 s at 0.025 m/s to 0.30 s at 0.16 m/s, with respective steady speed deficits of 2.59% and 0.38%. These quantify finite actuator response and transition demand. A speed ramp could reduce peak effort at the cost of slower departure, but that trade-off has not been experimentally compared; no improved tuning or energy saving is claimed.

Failure diagnosis and scope of robustness

Planner failure - seed 41. Both Q configurations stopped during Goal 3 planning. The later saved map establishes the immediate cause: an occupied cell 35.78 cm from the rounded robot cell lies within the 37.66 cm inflation radius. The robot cell can be observed free yet unsafe for its full footprint. This is a planning/map-clearance refusal, not demonstrated LQR instability. No camera update had been accepted; final localisation error was 11.56 cm. Pose drift and map discretisation are plausible upstream contributors, but the saved final snapshot does not identify the historical observation responsible. Reducing inflation might improve route availability while reducing clearance protection; no tested improvement is claimed.

Sensing failure — closer docking before the detector fallback. The noisy/slip trial reached approximately 1.91 cm actual dock error and 1.52° heading error, but marker detections ceased and three consecutive confirmations were never obtained before the operator closed the run. A static reconstruction reproduced detection failure even with the entire marker visible. An independent threshold-window fallback detected it and passed the existing PnP and EKF gates; the exact internal OpenCV candidate-selection mechanism was not established. The subsequent mode 3 demo completed camera-confirmed docking at 1.99 cm and 1.94° final true error without relaxing the docking thresholds. This supplies end-to-end evidence after the sensing change for this case, not a general detector success rate.

The earlier static detector sweep produced valid estimates at 97/105 poses (92.4%). Errors computed only over valid detections must be interpreted alongside that availability rate. A lower range RMSE does not compensate for complete measurement loss. The camera fallback addresses availability; retaining pose-quality checks and consecutive confirmations limits acceptance of unreliable observations. The earlier unsuccessful near-dock run remains a failed/interrupted trial rather than being retrospectively counted as successful.

Interruption tests address different mechanisms. The seed 41 Ctrl+C trial verified saving partial results while already stopped. The earlier two-second camera dropout verified continued prediction and reacquisition in an earlier noise configuration. Terminal pause/resume freezes simulation time and reset starts an independent trial with fresh state; neither demonstrates physical emergency braking or autonomous recovery from a planner stall. The latest demo adds successful operation under injected noise and physically measured slip, but final-configuration multi-seed repetitions would be needed to estimate condition-specific reliability.

Appendix

segment	samples	actual_rms_cm	estimated_rms_cm	actual_final_track_ey_cm	actual_final_track_heading_deg	actual_recovery_s	estimated_recovery_s	actual_endpoint_cm	last_track_time	actual_track_fraction_in_band
0	235	0.842842465	0.053798977	-1.136671507	-0.023906127	0.002	0.002	1.641878045	11.934	0.527659574
0	235	2.021393672	1.982299303	-1.051731867	-0.237507311	5.788	4.852	1.68507226	11.934	0.217021277
0	235	2.244483778	1.955850326	-1.231089718	0.182097723		4.488	1.755558752	11.934	0
0	236	0.867207611	0.450505514	-1.110318528	-0.112838999	0.752	0.752	1.65628197	11.984	0.508474576
0	236	1.040792038	0.416095425	-1.174220834	0.048789884	0.752	0.752	1.706878308	11.984	0.394067797
0	235	0.486530017	0.06782728	-0.509800453	-0.041970283	1.992	1.94	1.506377897	13.634	0.978723404
0	174	0.583627119	0.044814697	-0.918178855	-0.386342893	0.002	0.002	1.154091698	8.884	1
1	99	0.273536963	0.340983974	0.323364294	0.375595073	12.284	12.234	1.154091698	17.012	0.949494949
0	234	1.452658886	0.065685107	-2.013002711	-0.835490218	0.002	0.002	2.822365469	11.884	0.316239316
0	235	2.120546314	1.990726246	-1.737176713	-0.99329488		5.216	2.666875287	11.934	0.080851064
0	235	2.745729188	1.921400194	-2.250294662	-0.67895456		4.488	2.927169046	11.934	0
0	236	1.18888383	0.477717141	-1.682098247	-0.802388479	0.752	2.096	2.527404119	11.984	0.220338983
0	236	1.815983531	0.392280124	-2.320228952	-0.888354532		0.752	3.004407455	11.984	0.033898305
0	235	0.345273694	0.078276546	0.490683979	-0.330077199	1.94	1.992	1.757433746	13.634	0.982978723
0	172	1.16960838	0.074112573	-1.607236448	-0.407372107	0.002	0.002	0.85498303	8.784	0.430232558
1	99	0.624863813	0.305972903	-0.756232824	-0.520637288	12.134	12.284	0.85498303	16.908	0.95959596
0	235	0.226636857	0.058129644	0.139461762	0.324999419	0.002	0.002	1.811711589	11.934	1
0	236	1.926426448	1.966597315	0.267625056	0.099938132	4.956	4.8	1.818378367	11.984	0.593220339
0	236	2.027165729	1.958732116	-0.008713549	0.497182758	6.256	4.488	1.771037152	11.984	0.487288136
0	236	0.432929149	0.440616444	0.214326793	0.234821556	0.752	0.752	1.86329044	11.984	0.936440678
0	236	0.590959079	0.428817078	0.012035111	0.343354995	0.752	0.802	1.865684772	11.984	0.936440678
0	235	0.658943434	0.081026844	1.0775858	-0.269441097	1.94	1.992	2.131043662	13.584	0.70212766
0	172	0.295718044	0.027665354	-0.399748866	0.441392067	0.002	0.002	1.556567716	8.784	1
1	98	0.35923614	0.452488653	-0.392352577	-0.239645055	12.234	12.284	1.556567716	16.908	0.948979592